UNIVERSITI TEKNOLOGI MALAYSIA

SECI1143-02: PROBABILITY & STATISTICAL DATA ANALYSIS

PROJECT 1 : EXERCISE HABITS AND ITS IMPACTS

NAME	MATRIC ID
AHMAD ZIYAAD BIN MOHD ABBAS	A23CS0206
AFIF SHAQIR IRFAN BIN ARQAM	A23CS0204
IMAN ABADI BIN MOHD NIZWAN	A23CS0084
MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	A23CS0250

LECTURER'S NAME: DR NOORFA HASZLINNA BINTI MUSTAFFA

DATE OF SUBMISSION: 6 MAY 2024

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1.0 INTRODUCTION

Introduction

Physical activity plays a crucial role in maintaining overall health and well-being, particularly among people. However, in today's fast-paced world filled with various distractions, understanding the exercise habits of people has become increasingly important. Our group has embarked on a project aimed at investigating the frequency of weekly workouts among teenagers to gain insights into their exercise routines and behaviors.

Project Overview

To accomplish our objective, we designed a comprehensive survey using Google Forms. The survey aimed to collect data on various aspects of teenagers' exercise habits, with a particular focus on two key questions:

1. Frequency of Physical Activity: How many times do teenagers engage in physical activity in a typical week?
2. Duration of Workout Sessions: On average, how long does each workout session last?

By gathering responses to these questions, our goal is to gain a deeper understanding of how frequently teenagers participate in organized physical activity on a weekly average. Additionally, we aim to investigate potential factors that may influence their exercise habits, such as age, gender, lifestyle, and environmental factors.

Significance of the Study

The findings of this study hold significant implications for public health initiatives and youth-focused programs. Understanding exercise habits can provide valuable insights into their general health and well-being. By identifying patterns and trends in exercise behaviors, we can develop targeted interventions and programs aimed at promoting healthier lifestyles among people.

Moreover, with the increasing prevalence of sedentary lifestyles and the rise in health-related issues among people, such as obesity and mental health concerns, there is a growing need for effective strategies to encourage regular physical activity. By shedding light on exercise habits, we can contribute to the development of evidence-based interventions and policies aimed at improving their overall health outcomes.

Data Visualization
To present our findings effectively, we will utilize various visualization techniques, including:
- Bar charts
- Pie charts
- Dot plots
- Histograms
- Ogives
- Skeletal box plots
- Scatterplot

These visualizations will help us illustrate the distribution of responses and highlight any patterns or trends observed in people's exercise habits.

2.0 DATA COLLECTION

The methodology that our group used in this project to collect data is through conducting a survey through “Google Form”.The survey conducted consists some parts which are the personal information of the respondent,the frequency of their weekly workouts routine

No.	Question	Answer	Level of measurement
1	Gender	● Male ● Female	Nominal
2	Age group	● short answer (integer)	Ratio
3	On average, how many days per week do you engage in exercise?	● short answer(integer)	Ratio
4	How many minutes do you typically spend exercising per session	● short answer(integer)	Interval
5	How many minutes do you typically spend exercising per session	● short answer(integer)	Ratio
6	How many hours of sleep do you typically get per night	● short answer(integer)	Ratio
7	How many kilograms do you weight?	● short answer(integer)	Ratio
8	How many hours of sedentary behavior (inactive) do you engage in per day on average?	● short answer(integer)	Ratio
9	What is your resting heart rate? (beats per minute)	● short answer(integer)	Ratio
10	What type of exercise do you engage in most frequently	● Aerobic/cardiovascular (e.g., running,cycling) ● Strength training (e.g., weightlifting) ● Flexibility/stretching (e.g., yoga, Pilates) ● Mixed (combination of different types)	Nominal

11	What is the most significant improvement that you noticed since starting an exercise routine?	● Higher muscle mass ● Improved cardiovascular function ● Achievement of target weight ● Better sleep quality ● Boosts energy ● Improves mood	Nominal
12	On a scale of 1 to 5, how would you rate the intensity of your typical exercise session?	● 1: Low intensity ● 2: Moderate intensity ● 3: Moderate to high intensity ● 4: High intensity ● 5: Very high intensity	Ordinal
13	On a scale of 1 to 5, how motivated are you to engage in regular exercise?	● 1: Not motivated at all ● 2: Slightly motivated ● 3: Somewhat motivated ● 4: Very motivated ● 5: Extremely motivated	Ordinal
14	On a scale of 1 to 5, how satisfied are you with the results of your exercise routine in terms of overall health improvement?	● 1: Not satisfied at all ● 2: Slightly satisfied ● 3: Moderately satisfied ● 4: Very satisfied ● 5: Extremely satisfied	Ordinal
15	On a scale of 1 to 5, how would you rate the level of difficulty of maintaining your exercise routine?	● 1: Very difficult ● 2: Difficult ● 3: Neutral ● 4: Easy ● 5: Very easy	Ordinal

3.0 DATA ANALYSIS

3.1 RESPONDENT BIOGRAPHY

3.1.1 RESPONDENT'S GENDER

Gender	Frequency	Percentage
Male	36	58.1%
Female	26	41.9%
TOTAL	62	100%

Table 3.1.1.1 Frequency and Percentage of Respondent's Gender

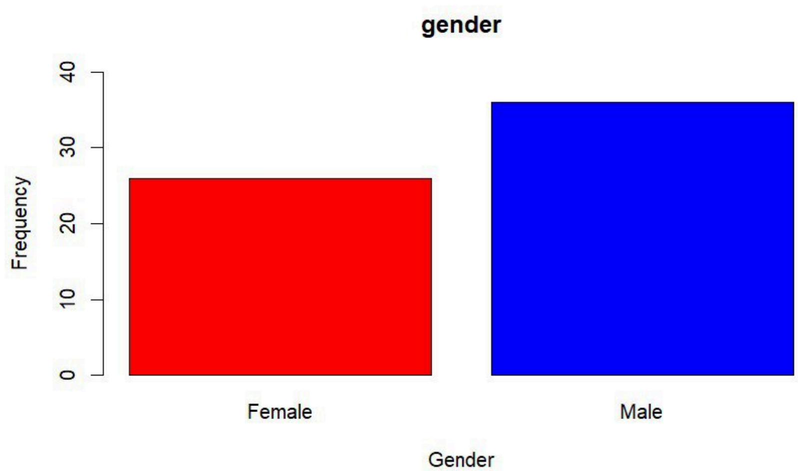


Figure 3.1.1.1 Histogram of Gender of Respondent

As indicated in Table 3.1.1.1 and Figure 3.1.1.1, which are the frequency table and histogram of respondent gender, the frequency of male is 36 and female is 26. As a result, male will have relative frequency, 0.58 while female is 0.42.The proportion is 58% for male and 42% for female.. There are 36 respondents from male and 26 respondents from female. In addition, the histogram of responder gender illustrates that red means female and blue means male.

3.1.2 Respondent's Age

Age	Frequency
15-20	41
20-25	15
25-30	2
30-35	3
35-40	1
TOTAL	62

Table 3.1.2.1 Frequency Distribution of Age of Respondents

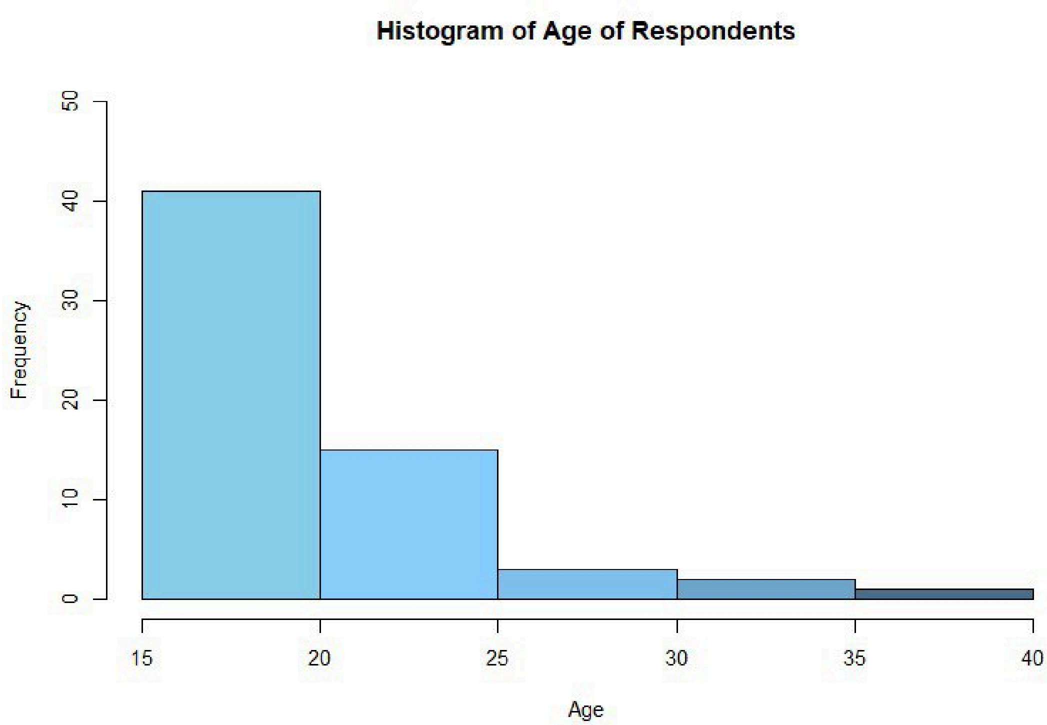


Figure 3.1.2.1 Histogram of Age of Respondents

Measure of Central Tendency	Value
Mean	20.08 years
Median	20-25 years
Mode	15-20 years

Table 3.1.2.2 Measure of Central Tendency of Age of Respondents

As seen in Table 3.1.1.2 and Figure 3.1.1.2,the age distribution of respondents that have answered our questionnaire are around group age of 15 to 40.Since this project is not specify on target age so it have many types of ages respondents.The highest frequency are in between age of 15 to 20 with the frequency of 41 because most of them are undergraduate students in Faculty of Computing.Students of age between 20 to 25 comes second with frequency of 14 and there are 3 respondents from age 30 to 35.Respondents from age 25 to 30 recorded frequency of 2 and the lowest frequency comes in the age of 35 to 40.

3.2 Health Metrics and Habits Assessment

3.2.1 Days engage in exercise per week

Days engage in exercise(per week)	Frequency
0	3
1	7
2	13
3	14
4	9
5	10
6	3
7	3
Total	62

Table 3.2.1.1 Frequency Distribution of Days engage in exercise(per week)

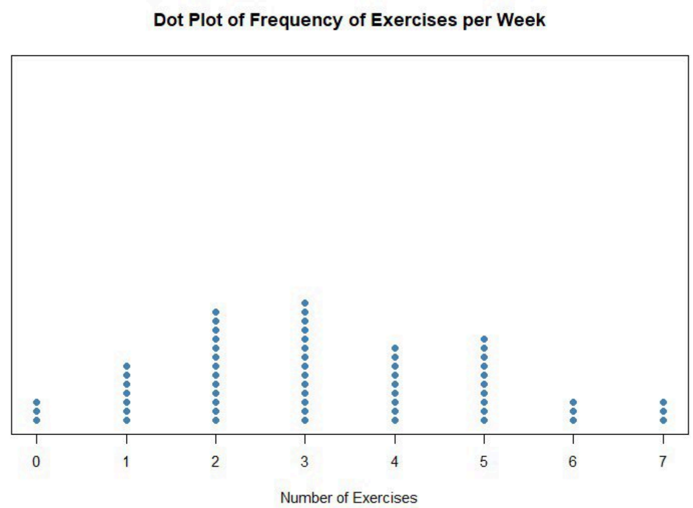


Figure 3.2.1.1 Dotplot of Days engage in exercise(per week)

In table 3.2.1.1 and Figure 3.2.1.1 shows the days respondent's engage in exercise per week.The lowest days respondent's engage in exercise per week are 0 days,6 days and 7 days with 3 respondents while the highest are 3 days per week with 14 respondents.The second highest frequency comes in 2 days of exercise per week with 13 respondents followed by 5 days of exercise per week with 10 respondents.The 4th highest frequency comes in 4 days of exercise with 9 days and 5th highest frequency are 1 days of exercise with frequency of 7.This means that most of the respondents will exercise regularly between 2 to 3 days and there are minimal frequency of respondents that do not exercise in a week.

3.2.2 Exercise duration spend by respondent's per session

Exercise Duration per session (minutes)	Frequency
0-20	14
20-40	23
40-60	17
60-80	1
80-100	1
100-120	6
Total	62

Table 3.2.2.1 Frequency Distribution of Exercise Duration per session (minutes)

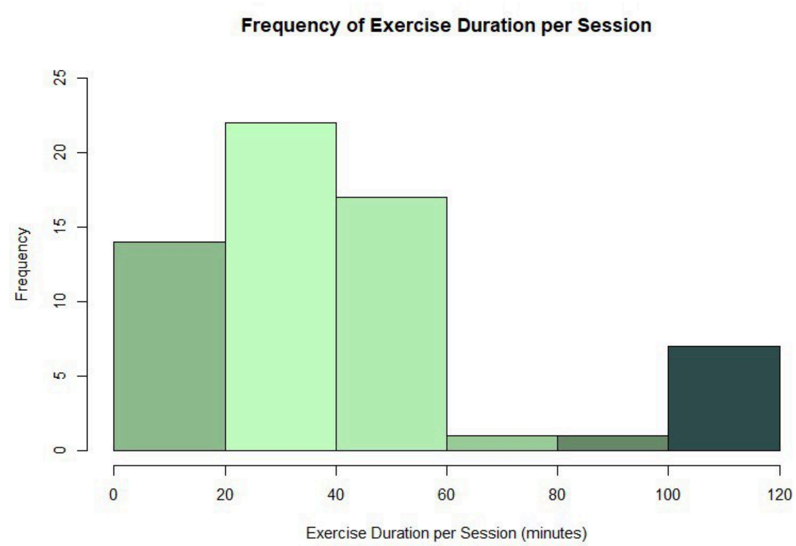


Figure 3.2.2.1 Histogram of Exercise Duration per session (minutes)

According to Table 3.2.2.1 and Figure 3.2.2.1,23 respondents indicated that they exercised for 20 to 40 minutes on average. This is clearly the majority of respondents. When the length goes above 40 minutes, there is noticeable reduction in frequency. There are a significant number of responders for both the 0–20 and 40–60 minute exercises—14 and 17, respectively. On the other hand, there is a noticeable decline in participation in sessions longer than sixty minutes. This indicates that although respondents' tastes vary, there appears to be a tendency toward moderately long exercise sessions.

3.2.3 Average Sleep Duration of Respondents

Sleep Duration (Hours)	Frequency
3	3
4	5
5	16
6	23
7	6
8	9
Total	62

Table 3.2.3.1 Frequency Distribution of Average Sleep Duration of Respondents (Hours)

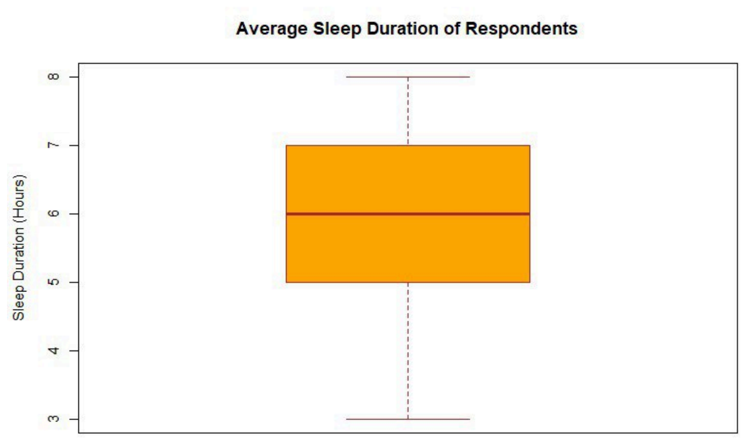


Figure 3.2.3.1 Skeletal Box Plot of Average Sleep Duration of Respondents (Hours)

Based on Table 3.2.3.1 and Figure 3.2.3.1, 23 respondents reported an average sleep length of 6 hours, which is the most common duration among the data. Nine respondents claimed eight hours of sleep, whereas 16 respondents reported five, symmetrically spread around this peak. Unexpectedly, only 3 people reported dangerous lengths, compared to 5 who claimed 4 hours. This indicates that getting about six hours of sleep is a typical goal, with differences on each side. Additional investigation into the variables affecting these patterns may provide insight into sleep patterns and their consequences for general health and wellbeing.

3.2.4 Respondent's Weight

Respondent's Weight	Frequency
40-50	10
50-60	20
60-70	10
70-80	13
80-90	6
90-100	2
100-110	1
Total	62

Table 3.2.4.1 Frequency Distribution of Weight of Respondents (Kilogram)

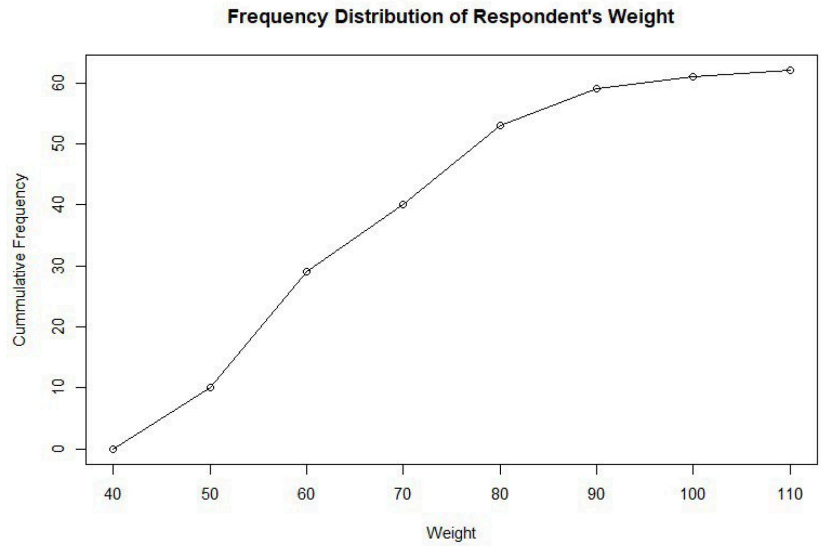


Figure 3.2.4.1 Ogive of Weight of Respondents (Kilogram)

Based on Table 3.2.4.1 and Figure 3.2.4.1,almost half of respondents fell between the 50 and 70 kg range (20 respondents in the 50-60 kg category and 13 in the 70-80 kg range). This suggests clustering in the middle of the weight distribution. But the frequency decreases as we approach the extremes, with just 2 respondents falling into the 90–100 kg bracket and just 1 responder into the 100–110 kg category. There are fewer people at the higher extremities of the weight spectrum, despite the fact that almost all of responses fell within a reasonably moderate weight range, according to this decreasing trend towards the higher weight groups.

3.2.5 Duration of Sedentary Behavior(hours) vs Resting Heart Rate (BPM)

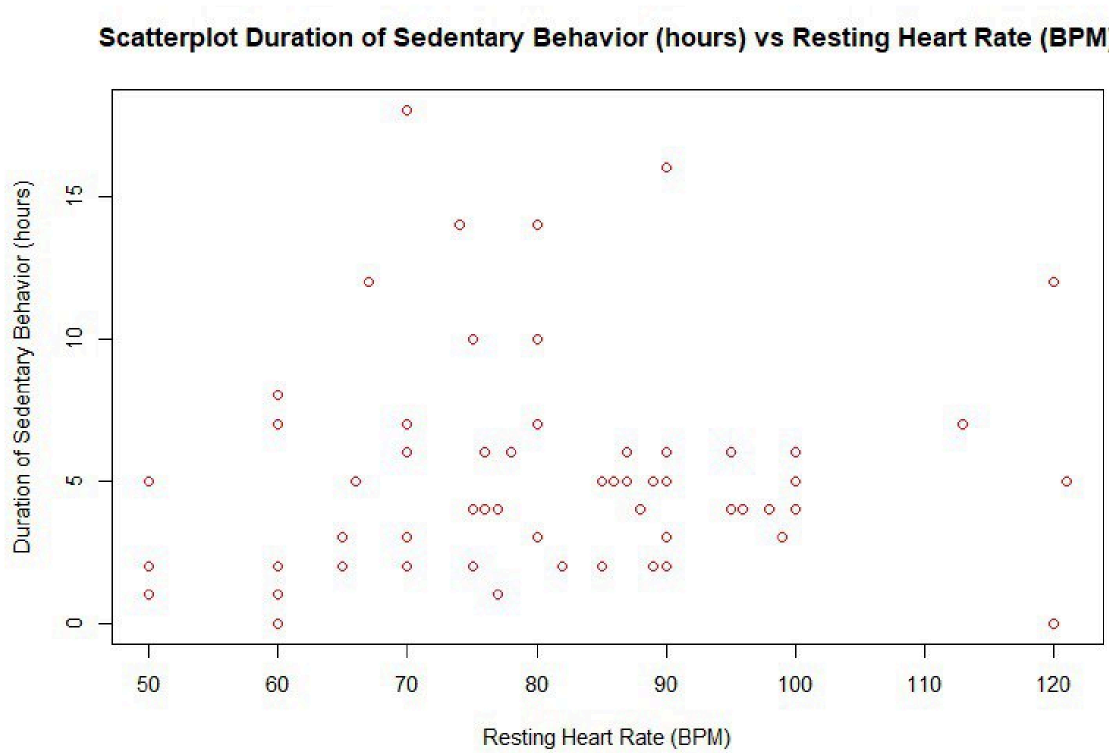


Figure 3.2.5.1 Scatterplot of Duration of Sedentary Behavior(hours) vs Resting Heart Rate (BPM)

When we look at a scatterplot comparing how much time someone spends sitting versus their resting heart rate, we can see if there's a connection between the two. If the points on the scatterplot generally go up and to the right, it suggests that when people sit for longer periods, their resting heart rate tends to be higher. According to (Jensen et al, 2013), higher resting heart rate was linked to lower physical fitness. This could mean that sitting a lot might not be great for heart health. Most of the sample holds true with the resting heart rate falling between the range of 60-100 BPM which is considered to be average while also having the duration of sedentary behavior to be less than 7. On the other hand, if the points don't follow a clear pattern, it could mean that sitting doesn't have much impact on resting heart rate. Understanding this relationship can help us figure out if sitting less might be good for keeping our hearts healthy.

3.2.6 Duration of Exercise Duration per Week (minutes) vs Resting Heart Rate (BPM)

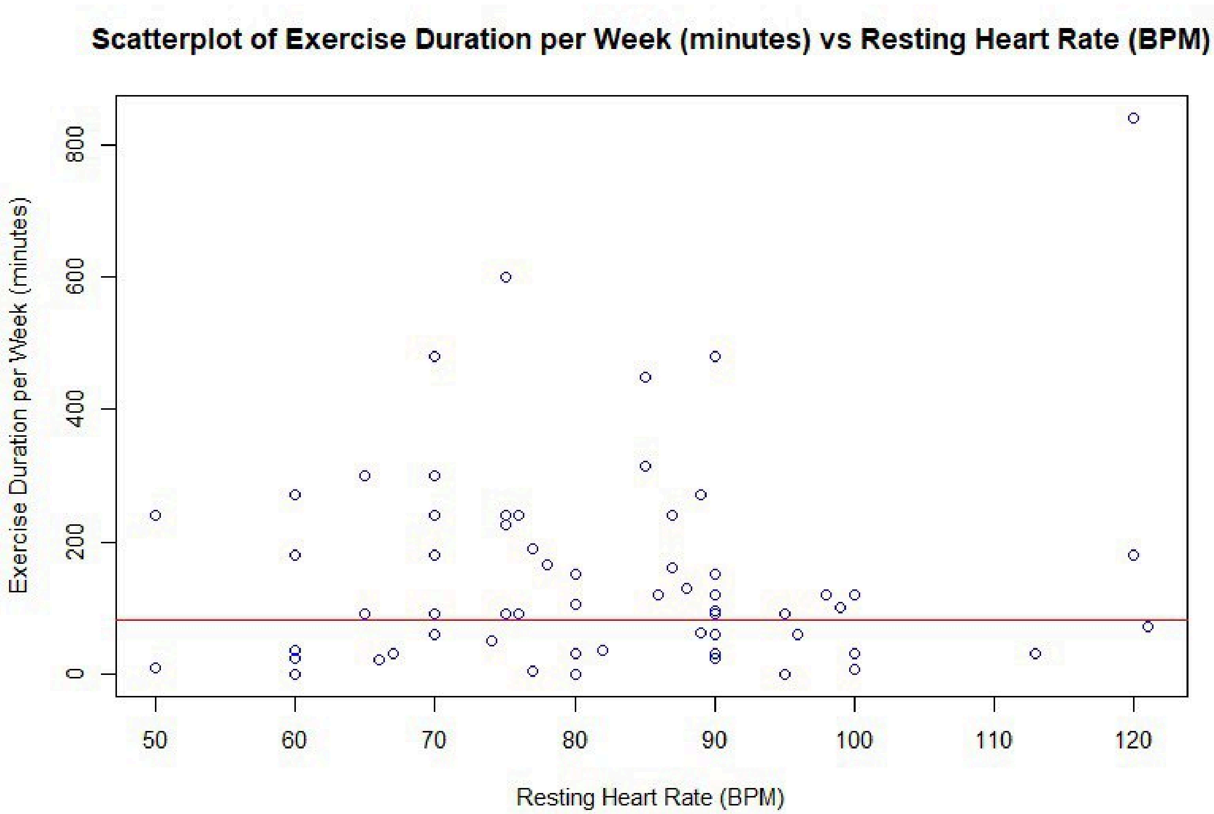


Figure 3.2.6.1 Scatterplot of Exercise Duration per Week (minutes) vs Resting Heart Rate (BPM)

Looking at a scatterplot comparing how much someone exercises each week versus their resting heart rate, we can see if there's a connection between the two. The value of exercise duration per week was calculated by multiplying the average time of a single exercise session with the number of exercise sessions per week. If the points on the scatterplot mostly go down as you move from left to right, it means that people who exercise more tend to have lower resting heart rates according to (Jensen et al, 2013) as well. This is a good sign for heart health because a lower resting heart rate usually means your heart is in better shape. Again most of the sample follow the trend, with only a few that do not. One extreme value in particular is the sample that has a resting heart rate of 120 BPM but does over 800 minutes of exercise per week. This raises questions if it could mean that exercise doesn't have a big impact on resting heart rate. Understanding this can help us figure out if exercising more might be good for keeping our hearts healthy.

3.3 Exercise Behavior and Perception Assessment

3.3.1 Exercise type

Type of exercises	Percentage
Aerobic/Cardiovascular training	30.6
Strength training	21
Flexibility/Stretching	16.1
Mixed	30.6
None	4.8

Table 3.3.1 Type of exercises (Percentage)

Figure 3.3.1 represents the distribution of different types of fitness services based on their usage rates. The most popular types of services are Aerobic Cardiovascular training and Mixed exercises, each accounting for 30.6% of the total. Strength training is the next most popular service, making up 21% of the total. Flexibility/Stretching exercises account for 16.1% of the services used. Interestingly, 4.8% of the respondents reported that they do not engage in any of the listed types of exercise. This data provides valuable insights into the exercise preferences of the surveyed group, with a clear preference for aerobic and mixed training. It's important to note that the percentages for 'Improves Mood' and 'Economic Benefits' are not specified in the table but are indicated to be less than 10%.

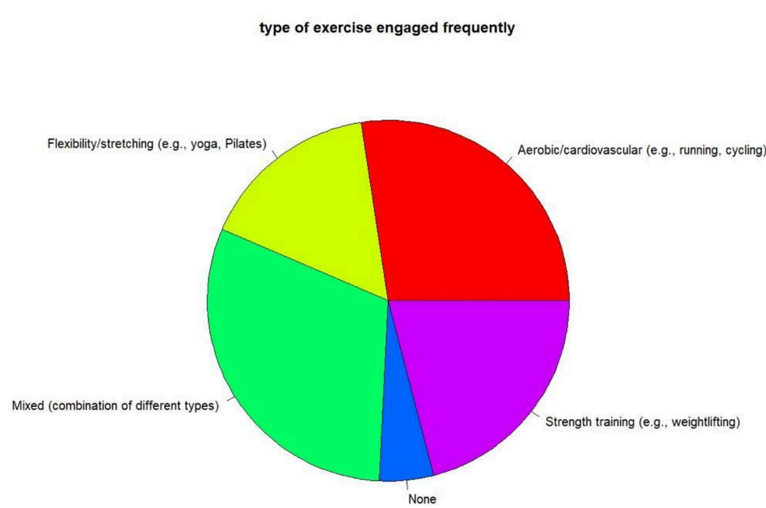


Figure 3.3.1 Pie Chart of Type of exercises (Percentage)

3.3.2 Significant improvement since starting exercise routine

Significant Improvement	Percentage
Higher Muscle Mass	17.7
Achievement of Target Weight	12.5
Boosts Energy	16.1
Improved Cardiovascular Function	12.9
Better Sleep Quality	11.3
Improves Mood	30.6

Table 3.3.2 Significant improvement since starting exercise routine (Percentages)

Table 3.3.2 lists several health-related improvements and their corresponding percentages. The most significant improvement is seen in mood, with a 30.6% increase, highlighting the emotional benefits of certain health practices. Higher muscle mass shows a 17.7% improvement, indicating a notable physical transformation. Boosts in energy and improved cardiovascular function are also evident, with 16.1% and 12.9% improvements respectively. Achieving target weight and better sleep quality are other key benefits, with 12.5% and 11.3% improvements. This data underscores the multifaceted benefits of health interventions, from physical to psychological well-being. The percentages provide a clear indication of the areas where individuals have experienced the most significant benefits.

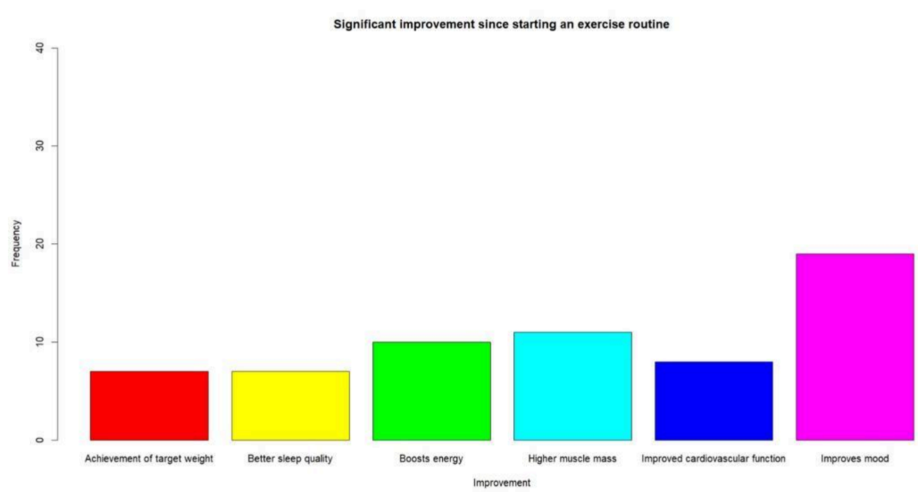


Figure 3.3.2 Pie Chart of Significant improvement since starting exercise routine (Percentages)

3.3.3 Exercise Intensity

Intensity	Percentage
1: Low intensity	12.9
2: Moderate Intensity	25.8
3: Moderate to high intensity	40.3
4: High intensity	11.3
5: Very high intensity	9.7

Table 3.3.3 Exercise Intensity (Percentages)

Table 3.3.3 outlines five levels of intensity, ranging from low to very high. The most common intensity level is ‘Moderate to High,’ accounting for 40.3% of the sample or population. ‘Moderate Intensity’ follows with 25.8%, suggesting a preference for moderate levels of activity. ‘Low Intensity’ and ‘High Intensity’ are less common, with 12.9% and 11.3% respectively. ‘Very High Intensity’ is the least common at 9.7%. This distribution indicates that while there is some engagement in both lower and higher intensity activities, most individuals tend to prefer a moderate to high level of intensity in their activities or conditions. The percentages provide insight into the preferred intensity levels within the group studied.

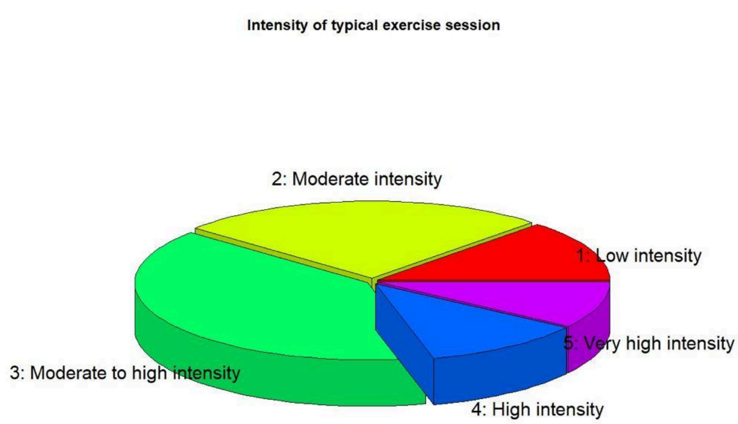


Figure 3.3.3 Pie Chart of Exercise Intensity (Percentages)

3.3.4 Exercise Motivation

Motivation	Percentage
1: Not motivated at all	6.5
2: Slightly motivated	22.6
3: Moderately motivated	35.5
4: Very motivated	24.2
5: Extremely motivated	11.3

Table 3.3.4 Exercise Motivation (Percentages)

Table 3.3.4 highlights five distinct levels of motivation among individuals. The majority, 35.5%, are 'Moderately motivated,' indicating a balanced level of motivation. 'Very motivated' individuals make up 24.2% of the group, showing a strong commitment to their goals. 'Slightly motivated' individuals account for 22.6%, suggesting a lower but present level of motivation. Those who are 'Not motivated at all' and 'Extremely motivated' represent the minority, with 6.5% and 11.3% respectively. This distribution suggests that while a significant portion of the population possesses a moderate to high level of motivation, there are still notable segments with lower or extremely high motivation levels. The percentages provide a quantitative insight into the motivational landscape of the group studied.

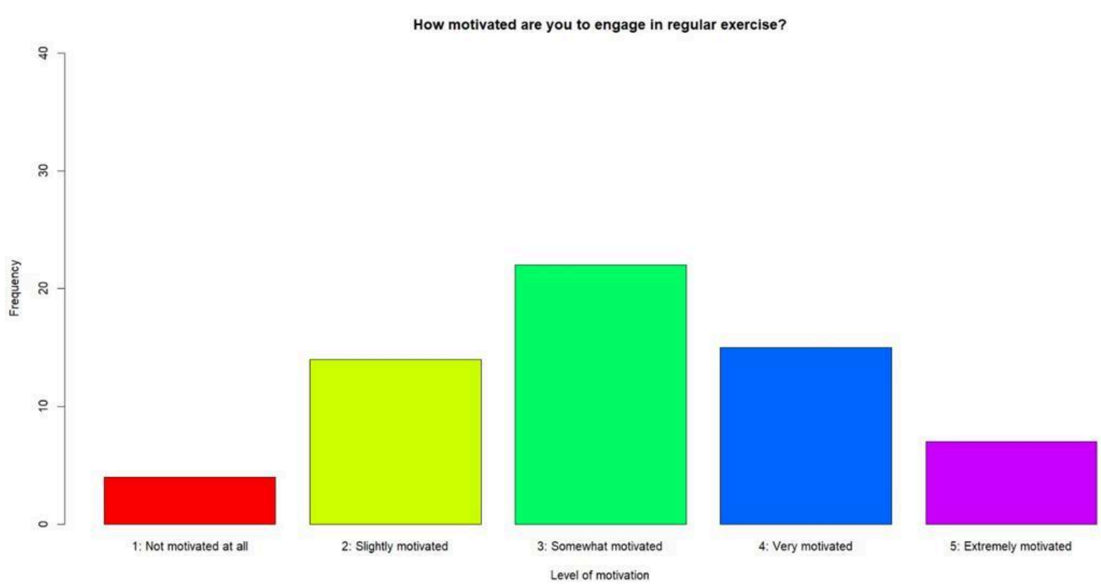


Figure 3.3.4 Pie Chart of Exercise Motivation (Percentages)

3.3.5 Exercise Satisfaction

Satisfaction	Percentage
1: Not satisfied at all	9.7
2: Slightly satisfied	22.6
3: Moderately satisfied	27.4
4: Very satisfied	32.3
5: Extremely satisfied	8.1

Table 3.3.5 Exercise Satisfaction (Percentages)

Table 3.3.5 displays five levels of satisfaction, ranging from 'Not satisfied at all' to 'Extremely satisfied.' The largest percentage of respondents, 32.3%, report being 'Very satisfied,' indicating a high level of contentment. 'Moderately satisfied' follows with 27.4%, showing that a significant portion of the group is fairly content. 'Slightly satisfied' individuals make up 22.6% of the group, suggesting some level of satisfaction. Those who are 'Not satisfied at all' represent 9.7%, while 'Extremely satisfied' individuals are the least common at 8.1%. This distribution provides insight into the overall satisfaction levels of the respondents, with most leaning towards the higher end of the satisfaction scale. The percentages offer a clear view of the group's sentiments regarding a service or product.

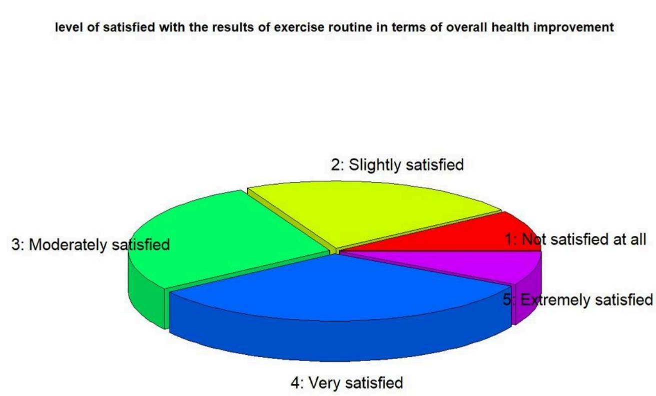


Figure 3.3.5 Pie Chart of Exercise Satisfaction (Percentages)

3.3.6 Exercise Routine Maintenance Difficulty

Difficulty	Percentage
1: Very difficult	6.5
2: Difficult	27.4
3: Neutral	43.5
4: Easy	16.1
5: Very Easy	6.5

Table 3.3.6 Exercise Routine Maintenance Difficulty (Percentages)

Table 3.3.6 delineates five levels of difficulty, from ‘Very difficult’ to ‘Very Easy.’ The majority of responses fall into the ‘Neutral’ category, comprising 43.5% of the total, indicating a balanced perception of difficulty. ‘Difficult’ is the next most common response, with 27.4%, suggesting that a significant portion of the respondents find the task or subject challenging. Both ‘Very difficult’ and ‘Very Easy’ are the least common responses, each with 6.5%, showing that extremes are less frequent. ‘Easy’ accounts for 16.1%, reflecting a moderate level of difficulty. This distribution of responses provides insight into how the task or subject is perceived in terms of difficulty by the respondents. The percentages offer a quantitative measure of the perceived challenge level within the group studied.

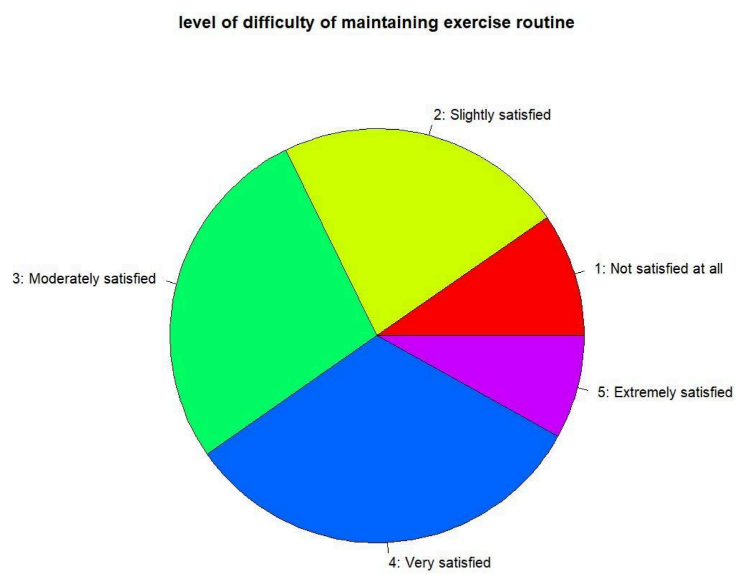


Figure 3.3.6 Pie Chart of Exercise Routine Maintenance Difficulty (Percentages)

4.0 CONCLUSION

In conclusion, the data presents a comprehensive overview of the respondents' health and fitness behaviors, motivations, satisfaction levels, and perceived difficulties. The majority of respondents are male, aged between 15 to 20, and engage in exercise 2 to 3 days per week for 20 to 40 minutes. The most common sleep duration is 6 hours, and the majority of respondents fall within the 50 to 70 kg weight range. The data also suggests a potential correlation between sitting duration and resting heart rate, as well as between exercise frequency and resting heart rate. However, these relationships require further investigation for confirmation.

The most popular fitness services among the respondents are Aerobic Cardiovascular training and Mixed exercises, indicating a preference for varied and cardiovascular-focused workouts. The data also highlights the multifaceted benefits of regular exercise, with improvements noted in mood, muscle mass, energy levels, cardiovascular function, weight management, and sleep quality. Finally, the respondents tend to prefer moderate to high intensity activities, suggesting a balance between challenge and comfort in their exercise routines. This comprehensive analysis provides valuable insights into the health and fitness behaviors and preferences of the respondents, which can be used to inform future health promotion strategies and interventions.

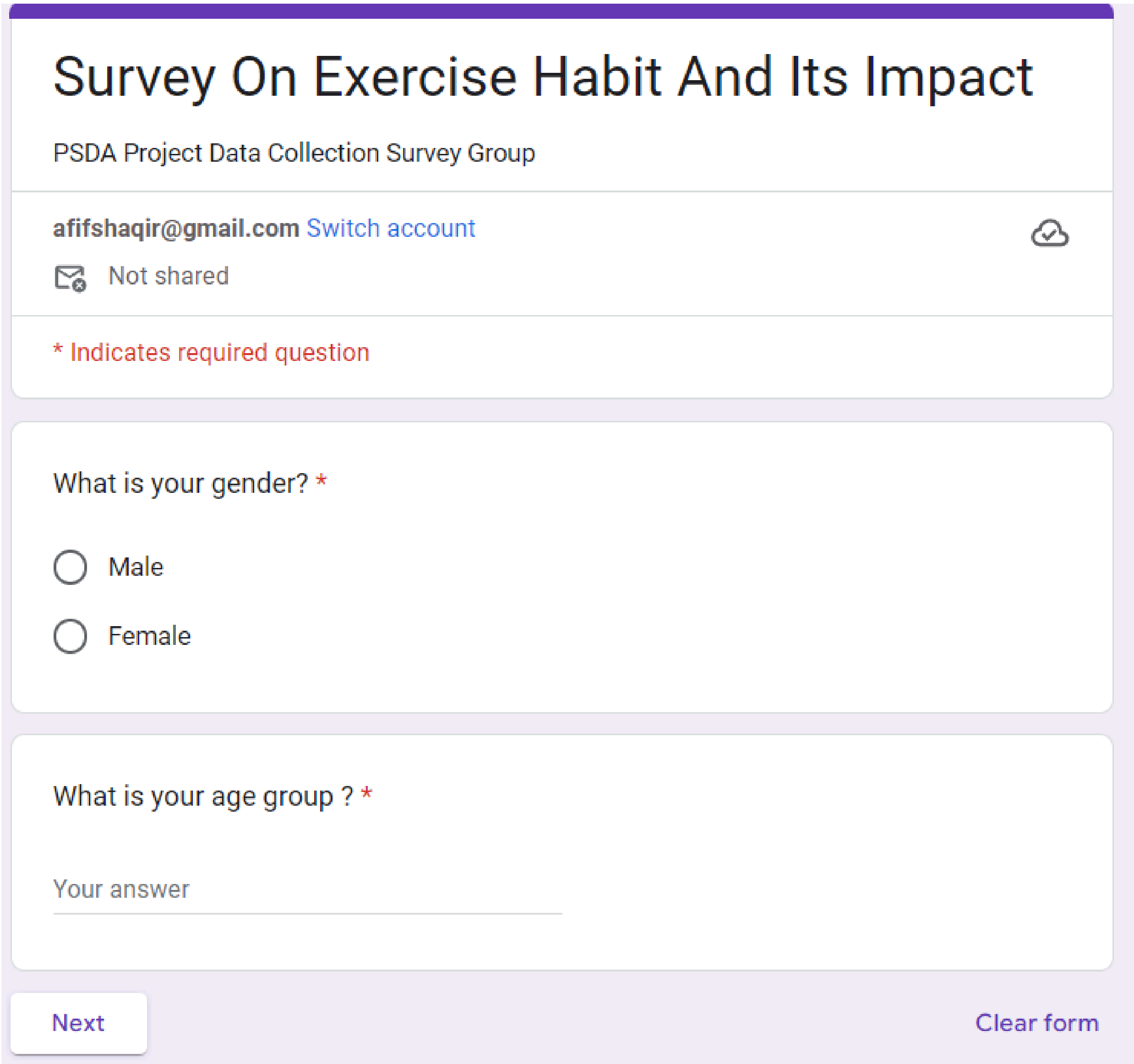
In terms of motivation, the majority of the respondents are moderately motivated, indicating a balanced level of motivation. However, there are still notable segments with lower or extremely high motivation levels. This suggests that while a significant portion of the population possesses a moderate to high level of motivation, there are still areas for improvement. The satisfaction levels of the respondents lean towards the higher end of the scale, with the majority reporting being very satisfied or moderately satisfied. This indicates a high level of contentment among the respondents, suggesting that the current health and fitness practices are meeting their needs and expectations to a large extent. The perceived difficulty of the health and fitness practices is mostly neutral, indicating a balanced perception of difficulty. However, a significant portion of the respondents find the practices challenging, suggesting that there may be a need for more personalized or adaptable practices to cater to different fitness levels and capabilities.

Overall, this comprehensive analysis provides valuable insights into the health and fitness behaviors, motivations, satisfaction levels, and perceived difficulties of the respondents. These insights can be used to inform future health promotion strategies and interventions, with the aim of improving health and fitness outcomes for the population. The data underscores the importance of a holistic approach to health and fitness, taking into account not only physical behaviors and outcomes, but also psychological factors such as motivation and satisfaction. By understanding and addressing these factors, we can create more effective and sustainable health and fitness practices.

5.0 REFERENCES

1. Jensen, M. T., Suadicani, P., Hein, H. O., & Gyntelberg, F. (2013). Elevated resting heart rate, physical fitness and all-cause mortality: a 16-year follow-up in the Copenhagen Male Study. *Heart*, *99*(12), 882–887. https://doi.org/10.1136/heartjnl-2012-303375

APPENDIX



Survey On Exercise Habit And Its Impact

PSDA Project Data Collection Survey Group

afifshaqir@gmail.com Switch account

Not shared

* Indicates required question

What is your gender? *

- Male
- Female

What is your age group ? *

Your answer

Next

Clear form

Health Metrics and Habits Assessment

On average, how many days per week do you engage in exercise? *

Your answer

How many minutes do you typically spend exercising per session *

Your answer

How many hours of sleep do you typically get per night *

Your answer

How many kilograms do you weight? *

Your answer

How many hours of sedentary behavior (inactive) do you engage in per day on average? *

Your answer

What is your resting heart rate? (beats per minute) *

Your answer

Exercise Behavior and Perception Assessment

What type of exercise do you engage in most frequently *

- ◯ Aerobic/cardiovascular (e.g., running, cycling)
- ◯ Strength training (e.g., weightlifting)
- ◯ Flexibility/stretching (e.g., yoga, Pilates)
- ◯ Mixed (combination of different types)
- ◯ None

What is the most significant improvement that you noticed since starting an exercise routine? (Nominal) *

- ◯ Higher muscle mass
- ◯ Improved cardiovascular function
- ◯ Achievement of target weight
- ◯ Better sleep quality
- ◯ Boosts energy
- ◯ Improves mood

On a scale of 1 to 5, how would you rate the intensity of your typical exercise session? *

- ◯ 1: Low intensity
- ◯ 2: Moderate intensity
- ◯ 3: Moderate to high intensity
- ◯ 4: High intensity

On a scale of 1 to 6, how motivated are you to engage in regular exercise? *

- 1: Not motivated at all
- 2: Slightly motivated
- 3: Somewhat motivated
- 4: Very motivated
- 5: Extremely motivated

On a scale of 1 to 5, how satisfied are you with the results of your exercise routine in terms of overall health improvement? *

- 1: Not satisfied at all
- 2: Slightly satisfied
- 3: Moderately satisfied
- 4: Very satisfied
- 5: Extremely satisfied

On a scale of 1 to 5, how would you rate the level of difficulty of maintaining your exercise routine? *

- 1: Very difficult
- 2: Difficult
- 3: Neutral
- 4: Easy
- 5: Very easy

Survey On Exercise Habit And Its Impact

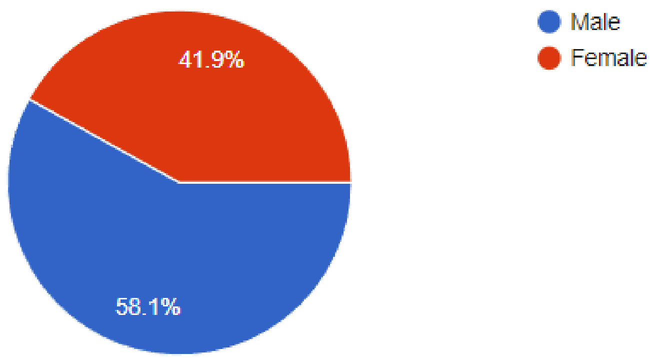
62 responses

Publish analytics

What is your gender?

Copy

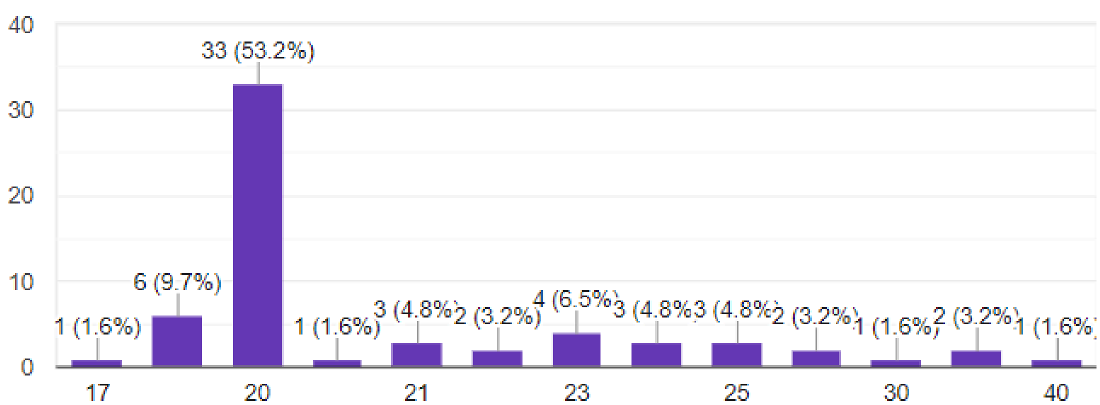
62 responses



What is your age group ?

Copy

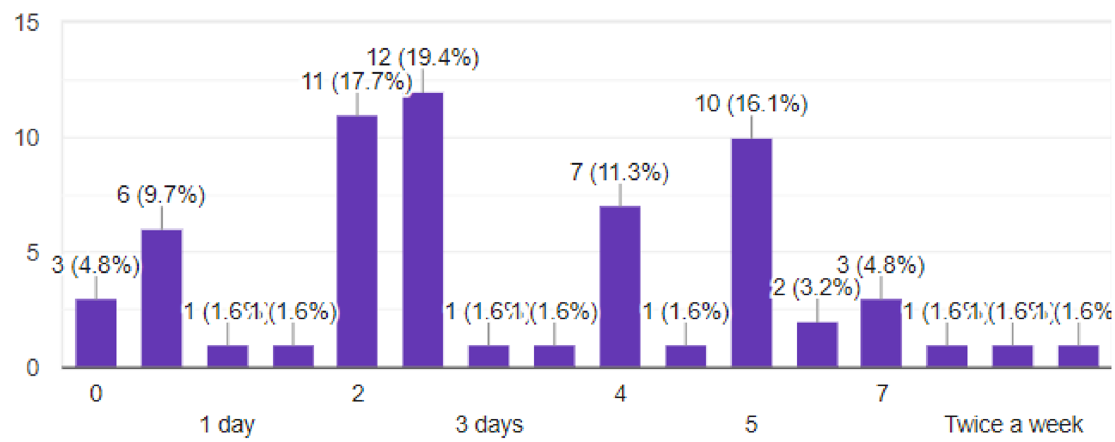
62 responses



Health Metrics and Habits Assessment

On average, how many days per week do you engage in exercise?

62 responses



How many minutes do you typically spend exercising per session

62 responses